

Palliative Care Intervention for Patients With End-Stage Liver Disease

A Cluster Randomized Clinical Trial

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IMPORTANCE Palliative care improves quality of life (QoL) in advanced illnesses, but data in end-stage liver disease (ESLD) are limited. It is unknown whether palliative care delivered by hepatologists is effective when compared with palliative care specialists.

OBJECTIVE To compare the effectiveness of palliative care delivered by trained hepatologists with the care delivered by conventional palliative care specialists in improving QoL at 3 months.

DESIGN, SETTING, AND PARTICIPANTS This comparative effectiveness cluster randomized trial for US patients treated for ESLD in 19 US medical centers compared a palliative care intervention delivered by palliative care-trained hepatologists (hepatologist group; 11 centers) with palliative care specialists (consultative group; 8 centers). Eligible patients were US adults with either decompensated cirrhosis or hepatocellular cancer who had a life expectancy of at least 6 months, had not received or scheduled liver transplantation, or had not received palliative care in the prior 3 months. Hepatologists in hepatologist group alone received primary palliative care training. Data collection occurred from January 2019 through June 2025; analysis was conducted from July to September 2025.

INTERVENTION Participants received 4 palliative care visits over 3 months delivered by either palliative care-trained hepatologists or palliative care specialists, using a structured palliative care checklist.

MAIN OUTCOMES AND MEASURES Superiority or a priori noninferiority of the effect of palliative care delivered by hepatologists vs palliative care specialists on change in QoL at 3 months, measured by the Functional Assessment of Cancer Therapy-Hepatobiliary (FACT-Hep) total score (higher scores indicating better QoL). Secondary outcomes included change in symptom burden, distress, depression, satisfaction from baseline to 3 months, and mortality.

RESULTS A total of 935 patients were enrolled (mean [SD] age, 63.0 [10.3] years; 275 female [29%]; 130 Hispanic ethnicity [14%]; 144 Black [15%], 736 White [79%]). From baseline to 3 months, QoL improved in both groups (adjusted mean: hepatologist, 8.01 [95% CI, 5.38 to 10.65]; consultative, 7.02 [95% CI, 4.34 to 9.71]; both $P < .001$). Although superiority was not found in change in QoL, prespecified noninferiority analysis showed that the improvement in the hepatologist group was noninferior to the consultative group (adjusted mean difference, 0.98 [95% CI, -2.86 to 4.83]; $P = .01$). Symptom burden (adjusted mean difference, -7.52 [95% CI, -9.89 to -5.15] vs -5.31 [95% CI, -7.60 to -3.03]) and depression (adjusted mean difference, -1.18 [95% CI, -1.78 to -0.57] vs -0.90 [95% CI, -1.49 to -0.31]) improved in both groups, without significant between-group differences. Patient satisfaction improved more in the hepatologist group compared with the palliative care group (adjusted mean difference, 3.37 [95% CI, 2.24 to 4.49] vs 0.91 [95% CI, -0.15 to 1.96]; $P = .002$). Mortality at 3 months was similar in both groups.

CONCLUSIONS AND RELEVANCE This cluster trial found that palliative care delivered by trained hepatologists was comparable with palliative care delivered by palliative care specialists in improving QoL in patients with ESLD and was associated with greater improvement in patient satisfaction, demonstrating the effectiveness among enrolled patients.

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End-stage liver disease (ESLD) is a progressive, life-limiting condition affecting more than 10 million people worldwide.^{1,2} Once decompensation occurs, 2-year mortality approaches 50%,^{3,4} and patients experience substantial symptom burden (such as fatigue, cachexia, pruritus, and cognitive and functional decline), which markedly impairs patients' quality of life (QoL) and places immense burden on health care systems.⁵⁻⁷ Impaired QoL is associated with decreased adherence to treatments, increased health care utilization, and worse survival.^{8,9} Although liver transplant offers potential cure, it is available to only a minority of patients, creating an urgent need for effective interventions that improve QoL and reduce symptom burden.^{10,11}

Palliative care improves QoL and reduces symptom burden for patients with serious illnesses by addressing physical symptoms and psychosocial, emotional, and spiritual needs, as reported in systematic reviews.^{12,13} These benefits limit clinical equipoise for a non-palliative care control group.¹⁴ Despite its demonstrated benefits, palliative care remains underutilized in ESLD, primarily due to a national shortage of palliative care specialists, persistent misperceptions that palliative care is end-of-life care, and limited palliative care training among hepatologists.¹⁵ Only 17% of patients removed from the liver transplant waiting list received palliative care consultation and 89% of these consults were within 72 hours of death, typically in an inpatient setting.¹⁶ Moreover, limited evidence of effectiveness of palliative care in the context of ESLD contributes to delayed or absent referrals, leaving many patients unsupported during earlier stages of their illness.¹⁷⁻¹⁹

The scope of palliative medicine has expanded beyond hospice to a more supportive approach.^{20,21} Advances in palliative care research and practice emphasize early integration, outpatient delivery, and the use of non-palliative care specialists to address basic palliative needs for advanced illnesses.^{22,23} Meeting the growing needs of patients with ESLD therefore requires hepatologists to incorporate primary palliative care skills into routine practice.^{24,25} Yet, structured palliative care interventions and rigorous comparative effectiveness research for hepatologist-delivered palliative care for ESLD are lacking.^{26,27} Accordingly, we designed a comparative effectiveness trial, an approach that evaluates alternative care models rather than a nontreatment control, thereby enhancing relevance to contemporary practice.

The purpose of this study was to examine whether the delivery of palliative care by trained hepatologists was comparable with conventional palliative care specialists in improving QoL among patients with ESLD.²⁸ Our primary hypothesis was that trained hepatologist-delivered palliative care is superior to specialist-delivered palliative care (consultative group) in improving QoL at 3 months (primary end point), with an a priori noninferiority analysis planned if superiority was not demonstrated.

Methods

Study Design

The PAL-LIVER (Palliative Care for Liver Diseases) trial was a cluster randomized trial conducted at 19 academic health care

Key Points

Question In patients with end-stage liver disease (ESLD), does palliative care delivered by trained hepatologists differ from that provided by palliative care specialists?

Findings In this cluster trial of 935 participants, palliative care delivered by trained hepatologists did not demonstrate superiority but was found to be noninferior to palliative care delivered by specialists in improving quality of life at 3 months.

Meaning Hepatologist-delivered palliative care achieved outcomes comparable with specialist-delivered palliative care, supporting 2 effective evidence-based models for early palliative care delivery for patients with ESLD; these findings allow health systems a choice for implementation, aligned with their staffing and workflows.

institutions (centers) within the US. The study protocol and all modifications were approved by a data coordinating center and each center's institutional review board ([Supplement 1](#); statistical analysis plan available in [Supplement 2](#)).²⁴ In this study, we report the primary outcome and key secondary outcomes. The study enrolled caregivers as well; caregiver outcomes will be reported separately. Clinical centers were recruited from the American Association for the Study of Liver Diseases (AASLD)'s special interest group focused on health services research, and through Veterans Health Administration's (VA) liver consortium. The centers had an understanding that if their respective center was randomized to the consultative group (model 1), then the hepatologist would not undergo any training and would align their workflows for palliative care intervention to be delivered by their local board-certified palliative care professionals.

Hepatologists in the hepatologist group (model 2) underwent 12 weeks of liver-specific palliative care training, conducted under the supervision of a palliative care clinician and educator, and moderated by a specialist. Data collection and management were centralized at the data coordinating center. We established a patient and caregiver advisory board that contributed lived experiences to the study's design and conduct. This study is reported in accordance with the Consolidated Standards of Reporting Trials ([CONSORT](#)) reporting guideline.

Participants

Eligible patients were adults (aged 18 years or older) with cirrhosis and a decompensation event (ascites, variceal bleeding, or hepatic encephalopathy) within the prior 6 months, or with hepatocellular cancer (HCC) (except Barcelona Stage D, due to life expectancy of less than 6 months). Additional inclusion criteria included English literacy and the capacity to complete study assessments. Exclusion criteria were life expectancy of less than 6 months (determined by the treating hepatologists' clinical judgment), prior liver transplant or anticipated transplantation in the following 3 months, inability to consent, or receipt of palliative care within the previous 3 months. Written or verbal informed consent was obtained prior to enrollment. Due to COVID-19 and limited in-person visits,

we modified our protocol to allow verbal consent as well. Centers were allowed either option based on the circumstances. Even after COVID-19 restrictions were lifted, verbal consent was still allowed; and was approved by both sponsor and institutional review boards.

Randomization and Masking

Centers were randomized into hepatologist group (11 centers) or consultative group (8 centers). Randomization initially allocated 7 centers to each group, with 5 centers added later (1 to the consultative group, 4 to the hepatologist group) due to COVID-19-related enrollment disruptions and slower accrual in the hepatologist group. Initial randomization was stratified by VA vs non-VA centers to account for differences in access to palliative care services. Randomization was conducted by the data coordinating center. All investigators remained masked to outcome data until the completion of the trial.

Study Intervention

Participants received a 4-visit intervention at initial, and at 1, 2, and 3 months, delivered by board certified hepatologists in the hepatologist group, and by board-certified palliative care specialists in the consultative group. In both groups, clinicians and specialists followed a standardized palliative care approach focused on symptom management, psychosocial support, and prognosis-related conversations based on established palliative care guidelines.²⁹ Description of the intervention is provided in the protocol. Most patients received all intervention visits from the same clinician in each group. Patients were enrolled from within the same practice, so mostly hepatologists had existing clinical relationships with patients, with some internal referrals.

The COVID-19 pandemic prompted study extensions and protocol amendments, all of which were approved by the study sponsor (PCORI) and participating centers' institutional review boards. These amendments permitted remote consent and telehealth visits. Centers used their existing telehealth platforms to deliver the intervention. Consistent with other successful palliative care trials, intervention fidelity was assessed using a post-visit checklist completed by all clinicians or specialists documenting topics addressed (eAppendix in [Supplement 3](#)).³⁰

Palliative Care Training

We developed a liver disease-specific palliative care training program, Palliative Care Always-Hepatology, based on a 12-module online oncology palliative care curriculum. The details of the training program are published.³¹ Briefly, each module was 3 to 4 hours, with a monthly virtual live session. Approximately, 1 to 3 hepatologists per site underwent training. Competency among hepatology clinicians was evaluated using pretraining and posttraining questionnaires in conjunction with live virtual practice sessions, which allowed assessment of knowledge acquisition and communication skills. Refresher trainings were provided annually. No training was provided to palliative care specialists.

Recruitment and Data Collection

Participants were recruited from January 2019 through September 2024, with database lock in June 2025. Patients were

identified through hepatology clinic schedules and clinician referrals. Research coordinators screened patients for eligibility, approached eligible individuals for informed consent, in accordance with institutional review board approved procedures. Following consent and collection of baseline data, study visits occurred at initial and 1, 2, and 3 months. Sociodemographic and clinical data were obtained from either electronic health records or participant self-report. Patients were compensated \$40 after each study visit.

Outcome Measures

The primary outcome was change in QoL from baseline to 3 months, measured using Functional Assessment of Cancer Therapy-Hepatobiliary (FACT-Hep) total score. The FACT-Hep is a validated 45-item questionnaire comprising 27 general and 18 liver disease-specific items, with higher scores indicating better QoL.^{32,33} The general component (FACT-G) includes 4 subscales: physical, social and family, emotional, and functional well-being. Liver disease-specific items (eg, stomach pain, anorexia, jaundice) comprise the Hepatobiliary Cancer Subscale (FACT-HCS). FACT-Hep total score combines the FACT-G and FACT-HCS (range of 0 to 180). Minimal clinically important difference (MCID) for the FACT-Hep total score is 8 or 9 points, and 2 or 3 points for each subscale; and has been shown to be responsive to psychosocial interventions in patients with advanced liver diseases.³⁴

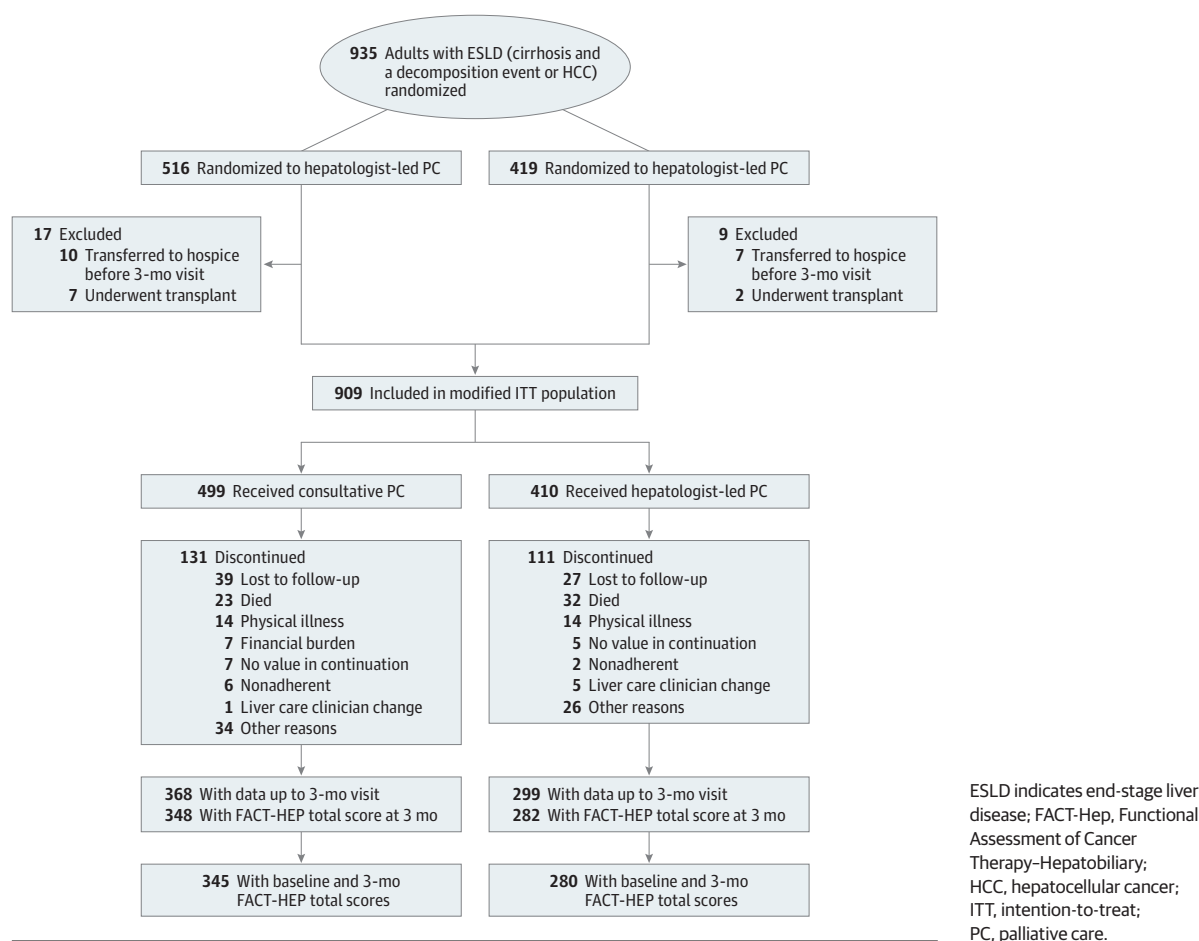
Secondary outcomes included change from baseline to 3 months in symptom burden measured using modified Edmonton Symptom Assessment Scale (ESAS),^{35,36} distress using Distress Thermometer (DT), depression using Patient Health Questionnaire-9 (PHQ-9), and satisfaction with care using Family Satisfaction with Care-Patient version (FAMCARE-P).³⁷ For all instruments, higher scores indicate greater levels of the domain assessed (eg, higher ESAS scores reflect greater symptom burden, whereas higher FAMCARE-P scores indicate higher patient satisfaction). Mortality was assessed at 3 months.

Statistical Analysis

The original design targeted enrollment of 1260 patients; however, due to COVID-19-related recruitment challenges, the sample size was reduced to 936 patients, ultimately 935 enrolled (1 excluded during data cleaning). Utilizing an estimated intraclass correlation coefficient and standard deviation of QoL change from interim study data by an independent unmasked statistical team, a sample size of 936 patients provided 83.2% power (based on simulations) to detect MCID of 9 points for FACT-Hep total score change.⁴⁴

In addition, before completion of the trial and with sponsor and data safety and monitoring board approval, we pre-specified a noninferiority hypothesis to be tested if the primary superiority comparison was nonsignificant. The rationale for incorporating noninferiority was to determine whether a trained hepatologist could provide care that was comparable with palliative care specialists, thereby addressing the clinical question of feasibility. The noninferiority margin was set at 4-point reduction in the mean FACT-Hep total score change in the hepatologist group (as compared with the consultative group) based on prior palliative care research.³⁸⁻⁴⁰ There was

Figure 1. Flow Chart of Enrollment, Randomization, and Follow-Up of Trial Patients



79.2% power for the noninferiority comparison, assuming the hepatologist group performed better than the consultative group by half of the MCID. A 2-sided significance level of .05 was considered for both superiority and noninferiority analyses.

All primary analyses were conducted in the modified intention-to-treat (ITT) population, comprising all enrolled patients excluding those who underwent liver transplant or transitioned to hospice before the 3-month assessment. These exclusions were prespecified, given these represent outcome-precluding events and follow a different clinical trajectory. All secondary outcomes were analyzed in an analogous way. Survival was evaluated in the ITT population.

For the primary analysis, to account for cluster randomization, we used a linear mixed-effects model with treatment arm indicator (fixed effect) and center as a random effect. Comparison of the primary outcome between the 2 groups was adjusted for the baseline FACT-Hep total score and VA status. To account for missing outcome data, we used multiple imputations with a fully conditional specification method.⁴¹ Multiple imputation used score values from baseline to 3 months, treatment assignment, VA status, and selected baseline demographics, with 1000 imputations generated for each analysis.

We performed 2 primary efficacy sensitivity analyses. We estimated survival with Kaplan-Meier curves and a *P* value

comparing 2 groups was obtained from a Cox proportional-hazards model with robust standard errors to account for the cluster randomization. We used SAS version 9.4 (SAS Institute) and R version 4.5.0 software (R Project for Statistical Computing).

Results

Study Participants

We enrolled 935 patients (mean [SD] age, 63.0 [10.3] years; 275 female [29%]; 130 Hispanic ethnicity [14%]; 144 Black [15%], 736 White [79%]), with 419 in the hepatologist group and 516 in the consultative group (Figure 1, Table 1). Baseline characteristics were largely comparable, except for a lower proportion of female patients in the consultative group (123 [24%]) compared with the hepatologist group (152 [36%]), reflecting a higher number of VA centers (with predominantly male populations) in the consultative group. The primary etiologies included alcohol-associated liver disease (322 [39%]), metabolic dysfunction-associated steatohepatitis (MASH) (230 [28%]), and viral hepatitis (196 [24%]). Baseline disease severity was similar in the hepatologist and consultative arms (mean [SD] Child-Pugh score, 8.6 [2.0] vs 8 [1.7]; mean [SD]

Table 1. Patient Characteristics in Intention-to-Treat Population (ITT)

Characteristic	Patients, No. (%)	
	Consultative palliative care (n = 516)	Hepatologist-led palliative care (n = 419)
Age, mean (SD) [range], y	64 (10.0) [25-87]	62 (10.4) [21-84]
Sex		
Female	123 (24)	152 (36)
Male	393 (76)	267 (64)
Hispanic or Latino ethnicity	62 (12)	68 (16)
Race		
African American or Black	85 (17)	59 (14)
White	406 (79)	330 (79)
Other ^a	23 (4)	28 (7)
Education		
High school diploma or less	255 (50)	198 (48)
Some college or bachelor's	244 (47)	195 (47)
Master's or above	16 (3)	23 (5)
Employment status		
Employed	59 (12)	71 (17)
Retired	248 (48)	163 (39)
On disability	136 (26)	77 (19)
Not employed	73 (14)	106 (25)
Marital status		
Single	122 (24)	105 (25)
Married	253 (49)	213 (51)
Divorced	107 (21)	65 (16)
Widowed	34 (6)	34 (8)
Insurance status		
No insurance	5 (1)	44 (11)
Private	157 (30)	140 (33)
Medicare	272 (53)	181 (43)
Medicaid	122 (24)	82 (20)
VA benefit insurance	222 (43)	62 (15)
Other insurance	6 (1)	41 (10)
Smoking status		
Never smoked	169 (33)	156 (37)
Former smoker	211 (41)	176 (42)
Current smoker	136 (26)	85 (20)
Duration of liver disease, y		
≤5	291 (63)	250 (66)
6-10	92 (20)	68 (18)
10-20	79 (17)	62 (16)
Primary liver diagnosis		
Alcohol-associated liver disease	178 (39)	144 (39)
MASH	112 (24)	118 (32)
Viral hepatitis (HCV or HBV)	122 (27)	74 (20)
Autoimmune hepatitis	12 (3)	14 (4)
Other (cryptogenic, A1AT, etc)	35 (8)	23 (6)
Presence of hepatocellular cancer at baseline	162 (31)	101 (24)
HCC stage		
A	46 (29)	14 (15)
B	71 (46)	29 (30)
C	39 (25)	53 (55)
MELD 3.0 score ≥20	57 (13)	63 (18)
Psychiatric issues		
History of depression	121 (26)	111 (30)
History of anxiety	59 (13)	74 (20)
Other (PTSD, bipolar, etc)	113 (24)	51 (14)

(continued)

Table 1. Patient Characteristics in Intention-to-Treat Population (ITT) (continued)

Characteristic	Patients, No. (%)	
	Consultative palliative care (n = 516)	Hepatologist-led palliative care (n = 419)
CCI score, mean (SD) [No. of patients] ^b	5.2 (2.9) [363]	4.5 (2.6) [376]
Caregiver co-enrollment	310 (60)	249 (59)
ECOG Performance Status scale ^c		
0	31 (9)	70 (19)
1	214 (61)	202 (54)
2	77 (22)	72 (19)
3	28 (8)	25 (7)
4	1 (<1)	2 (1)
MELD score 3.0		
Mean (SD)	14.2 (5.1)	15.0 (5.6)
Presence of hepatic encephalopathy	202 (45)	166 (44)
Presence of ascites	276 (62)	265 (71)
CTP score		
Mean (SD)	8 (1.7)	8.6 (2.0)
Class A	88 (20)	53 (14)
Class B	278 (62)	197 (53)
Class C	82 (18)	124 (33)
Baseline outcome measures, mean (SD)		
FACT-Hep total score (QoL)	118.2 (28.0)	113.4 (28.1)
ESAS-13 score (Symptom burden)	44.2 (25.4)	50.3 (27.1)
DT (Distress)	3.6 (3.0)	4.3 (3.0)
PHQ-9 (Depression)	8.0 (6.0)	10.0 (6.7)
FAMCARE-P (Satisfaction)	53.5 (10.0)	55.0 (9.6)

Abbreviations: A1AT, alpha-1 antitrypsin; CCI, Charlson Comorbidity Index; CTP, Child Turcotte Pugh; DT, Distress Thermometer; ECOG, Eastern Cooperative Oncology Group scale; ESAS-13, Edmonton Symptom Assessment Scale; FACT-Hep, Functional Assessment of Cancer Therapy-Hepatobiliary; FAMCARE-P, Family Satisfaction With Care-Patient version; HBV, hepatitis B virus; HCC, hepatocellular cancer; HCV, hepatitis C virus; MASH, metabolic dysfunction-associated steatohepatitis; MELD, Model for End Stage Liver Diseases; PHQ-9, Patient Health Questionnaire; PTSD, posttraumatic stress disorder; QoL, quality of life; VA, Veterans Administration.

^a Other included American Indian or Alaska Native, Asian, Native Hawaiian or another Pacific Islander, and multiple races.

^b CCI measure was age adjusted with liver disease excluded.

^c ECOG Performance Status Scale is a clinician-assessed measure of patients' functional status: 0, fully active with no restrictions; 1, able to do light work; 2, unable to work and in bed less than 50% of the day; 3, capable of only limited self-care and in bed more than 50% of the day; 4, bedridden.

MELD 3.0, 15.0 [5.6] vs 14.2 [5.1]; presence of HCC, 101 [24%] vs 162 [31%]). However, proportions of patients with HCC stage C and CTP class C were higher in the hepatologist group. At baseline, the FACT-Hep total score was slightly lower in the hepatologist than the consultative arm (mean [SD], 113.9 [28.1] vs 118.0 [28.0]; adjusted for in the primary analysis) (eTable 1 in Supplement 3).

Intervention Delivery

Among the 909 patients in the modified ITT population, 90% received an initial palliative care visit, with similar uptake across study groups (eTable 2 in Supplement 3). A higher proportion of patients in the hepatologist group completed the initial visit in-person compared with the consultative group (259

Table 2. Adjusted Mean Change in Primary and Secondary Outcomes at 3 Months in Modified Intention-to-Treat (ITT) population

	Adjusted mean (95% CI) ^a		Adjusted mean difference (95% CI) ^a
Outcome (change from baseline to 3 mo)	Consultative palliative care (model 1) (n = 499)	Hepatologist-led palliative care (model 2) (n = 410)	
Primary outcome			
FACT-Hep total score	7.02 (4.34 to 9.71)	8.01 (5.38 to 10.65)	0.98 (−2.86 to 4.83)
<i>P</i> value	<.001	<.001	.62
<i>P</i> for noninferiority ^b	NA	NA	.01
FACT-Hep subscales ^c			
PWB	1.18 (0.41 to 1.94)	1.39 (0.65 to 2.13)	0.21 (−0.88 to 1.30)
SWB	0.89 (0.22 to 1.57)	0.45 (−0.21 to 1.11)	−0.44 (−1.40 to 0.52)
EWB	0.63 (0.11 to 1.15)	1.46 (0.92 to 1.99)	0.83 (0.06 to 1.59)
FWB	1.06 (0.33 to 1.79)	1.26 (0.52 to 1.99)	0.20 (−0.87 to 1.26)
FACT-Hepatobiliary subscale (HCS)	3.50 (2.40 to 4.60)	3.28 (2.19 to 4.37)	−0.22 (−1.80 to 1.35)
Secondary outcomes ^c			
FACT-G (general) ^d	3.56 (1.68 to 5.43)	4.59 (2.72 to 6.45)	1.03 (−1.69 to 3.75)
FACT-TOI (trial outcome index) ^e	5.65 (3.52 to 7.78)	5.88 (3.79 to 7.96)	0.23 (−2.81 to 3.27)
Modified ESAS total score (13 items) ^f	−5.31 (−7.60 to −3.03)	−7.52 (−9.89 to −5.15)	−2.21 (−5.58 to 1.16)
ESAS total score (10 items)	−3.83 (−5.50 to −2.15)	−5.36 (−7.14 to −3.59)	−1.53 (−4.03 to 0.95)
PHQ-9	−0.90 (−1.49 to −0.31)	−1.18 (−1.78 to −0.57)	−0.28 (−1.15 to 0.58)
DT	−0.34 (−0.63 to −0.06)	−0.27 (−0.58 to 0.03)	0.07 (−0.36 to 0.50)
FAMCARE-P 13	0.91 (−0.15 to 1.96)	3.37 (2.24 to 4.49)	2.46 (0.91 to 4.01)

Abbreviations: DT, Distress Thermometer; ESAS, Edmonton Symptom Assessment Scale; EWB, emotional well-being; FACT-Hep, Functional Assessment of Cancer Therapy–Hepatobiliary; FAMCARE-P, Family Satisfaction with Care–Patient; FWB, functional well-being; PHQ-9, Patient Health Questionnaire; PWB, physical well-being; SWB, social well-being.

^a Results are based on a random effect (site) model adjusted for the baseline score value and Veterans Administration (VA) status. Decrease in ESAS, DT, and PHQ-9 is improvement in symptoms, distress and depression; while an increase in FACT-Hep scores, FAMCARE-P is improvement in quality of life, and satisfaction with palliative care.

^b Noninferiority is established using the 2-sided 95% CI for the estimated

difference between groups (within the prespecified margin of −4 points) for hepatologist-led model being better than the consultative model.

^c The minimally clinically important differences for FACT-Hep scores are: FACT-Hep subscales, 2-3; FACT-G, 6-7; FACT-Hep total score, 8-9; HCS, 5-6; FACT-TOI, 7-8.⁴⁴

^d FACT-G = PWB + SWB + EWB + FWB.

^e FACT-TOI = PWB + FWB + HCS.

^f Modified ESAS total score range (13 items), 0-130. Higher scores reflect more symptom burden.

[70%] vs 255 [57%]). Approximately two-thirds of patients in both groups completed all 4 intervention visits (hepatologist, 242 [65%]; consultative, 286 [64%]). The intervention was delivered primarily in the outpatient setting; inpatient visits were uncommon (108 [13%]) and occurred only when a patient was hospitalized at the time of their study intervention visit. Cumulative fidelity scores calculated as the sum of weighted scores on the palliative care checklist (100-point scale, with each item scored once), and the 3-month total taken as the cumulative fidelity score) were high and comparable in both arms (mean [SD] scores: hepatologist, 92.3 [10.4]; consultative, 89.3 [12.8]). The attrition rates at 3 months were similar in both arms, primarily due to death, physical illness, or loss to follow-up.

Primary Outcome

Quality of life (FACT-Hep total score) improved significantly from baseline to 3 months in both comparators: an increase of 8.01 points (95% CI, 5.38 to 10.65; $P < .001$) in the hepatologist group, and 7.02 points (95% CI, 4.34 to 9.71; $P < .001$) in the consultative group (Table 2, Figure 2A). Adjusted difference between the 2 groups was not significant (0.98 [95% CI, −2.86 to 4.83]; $P = .62$). However, change in the hepatologist group was noninferior to the consultative group (P for noninferiority = .01). Intraclass correlation coefficient was 0.015. Two sensitivity analyses excluding patients who died prior to 3-month follow-up, and imputing low FACT-Hep

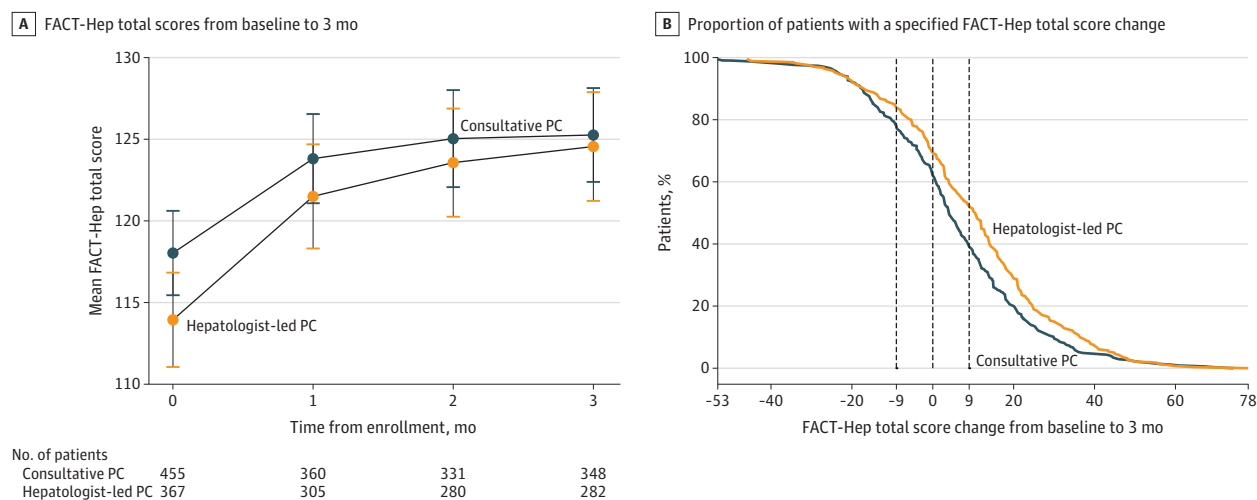
scores in those who died, yielded consistent results, confirming noninferiority of the hepatologist group (eTable 3 in Supplement 3). In a post hoc descriptive analysis, we summarized proportion of patients in each group who achieved a clinically meaningful improvement of 9 points or more (52% in the hepatologist group vs 39% in the consultative group) (Figure 2B).

Among the FACT-Hep subscales, change in emotional well-being (EWB) subscale suggested a greater improvement in the hepatologist group than in the consultative group at 3 months (0.83 [95% CI, 0.06 to 1.59]). The FACT-Hep HCS improved in both groups, with no difference between the 2 groups (−0.22 [95% CI, −1.80 to 1.35]). All other subscales demonstrated a similar pattern across both groups.

Secondary Outcomes

Secondary outcomes were comparable across arms, with modestly better results in the hepatologist group (eFigure 1 in Supplement 3; Table 2). The modified 13-item ESAS total score decreased by −7.52 points (95% CI, −9.89 to −5.15) in hepatologist group and −5.31 points (95% CI, −7.59 to −3.02) in consultative group with no difference between arms (2.21; 95% CI, −5.58 to 1.16). Depression scores (PHQ-9) also improved in both arms, by −1.18 points in the hepatologist group and −0.9 points in consultative group, with no difference between arms (−0.28; 95% CI, −1.15 to 0.58).

Figure 2. Line Graph of FACT-Hep Scores and Score Change in the 2 Comparative Models



A, Lines follow means and vertical bars are 95% CIs at each time point. B, Results are shown for the hepatologist and consultative groups based on a post hoc descriptive analysis using complete data. No formal statistical comparisons were performed. The proportion of patients with a change in

FACT-Hep score greater than 9 was 52% in the hepatologist group and 39% in consultative group. FACT-Hep indicates Functional Assessment of Cancer Therapy–Hepatobiliary; PC, palliative care.

Notably, patient satisfaction improved more in the hepatologist group (3.37 points) compared with the consultative group (0.91 points), with a between-arm difference of 2.46 (95% CI, 0.91 to 4.01; $P = .002$). Mortality was similar (eFigure 2 in Supplement 3). There were no statistically significant differential treatment effects across subgroups (eTable 4 in Supplement 3).

Discussion

In the PAL-LIVER trial, we found that among patients with ESLD, palliative care intervention delivered by trained hepatologists was noninferior to palliative care specialists in improving QoL at 3 months. This finding is notable given the natural history of ESLD, which is typically characterized by progressive symptom burden and declining QoL.^{42–45} In addition, patients receiving palliative care from trained hepatologists experienced greater improvements in emotional well-being and satisfaction with care, while improvements in symptom burden were comparable in both arms. Together, these findings underscore the importance of palliative care as a core component of ESLD management and demonstrate that hepatologists, with training in primary PC, can deliver effective primary palliative care for selected patients.⁴⁶

The PAL-LIVER intervention complemented routine hepatology care. The concurrent delivery of supportive care alongside active management of cirrhosis-related symptoms and procedures by hepatologists may have influenced patient perceptions of better emotional well-being, although the relative contribution of clinical continuity vs intervention content cannot be distinguished in this study.

Differences in mode of delivery are unlikely to account for the greater improvement in patient satisfaction observed in the

hepatology group, as a higher proportion of visits in consultative group were conducted in person (eTable 2 in Supplement 3). Satisfaction may instead reflect factors such as continuity with the primary liver care providers, established patient-clinician relationships, or patient expectations, which were not directly assessed in this trial.⁴⁷

Several additional factors may explain these findings. First, our results are consistent with prior evidence from other chronic, symptomatic illnesses demonstrating that palliative care interventions improve QoL.^{48–50} Extending this evidence to ESLD is particularly important, as patients with cirrhosis have historically been underrepresented in palliative care trials despite substantial symptom burden and psychological distress. Second, hepatologists may have leveraged established therapeutic relationships with ESLD patients, along with deeper familiarity with disease-specific symptoms, prognostic uncertainty, and illness trajectory. These factors may facilitate trust, continuity, and effective communication around symptoms and coping with illness. Third, to our knowledge, this is the first large-scale randomized trial to evaluate the effectiveness of hepatologist-delivered palliative care compared with specialty palliative care. Prior trials demonstrating improvements in QoL have largely relied on palliative care specialists and their teams.^{51,52} Our findings suggest that primary palliative care delivery by hepatologists may be particularly relevant in health systems with limited access to specialty palliative care.^{53,54}

Strengths and Limitations

We demonstrated that hepatologists can be trained to deliver palliative care associated with meaningful patient outcomes. The intervention was informed by implementation science principles and designed to be pragmatic, including adaptations for telehealth delivery. This approach aligns with national

recommendations calling for earlier and more systematic integration of palliative care into the care of patients with ESLD.^{46,47} System-level strategies could further support implementation of hepatologist-led palliative care.⁵⁵

Several limitations warrant consideration. First, although we enrolled a large and geographically diverse cohort, differences in recruitment between study groups potentially introduced selection bias. Lower enrollment in the hepatologist group may have been influenced by clinician availability, particularly limited hepatologist time for study intervention visits and workflow constraints such as restricted office hours and limited clinic space for longer palliative care visits. These challenges were further exacerbated by COVID-19 disruptions impacting in-person visits and staffing. Although attrition rates were similar between groups, lower enrollment in hepatologist group has important implications for external validity, scalability, and future adoption in routine practice. Notably, we enrolled English-speaking patients only.

An important alternative interpretation of our findings is that, despite noninferior outcomes, referral to palliative care specialists may represent an easier strategy for improving access to palliative care. Time, training, and workflow demands required to equip hepatologists to deliver palliative care may limit uptake in routine practice. Future implementation studies should directly compare not only clinical outcomes but also feasibility, reach, clinician burden, cost-effectiveness, and sustainability of trained hepatologist model vs palliative care specialist's referral.

Our qualitative interviews suggested that practice-based learning through delivering palliative care led to increased proficiency among hepatologists over time, reflecting experiential learning in delivering palliative care, although a longer study duration may be needed to fully understand the impact of learning curves on recruitment and delivery.⁵⁶

Second, despite randomization, patients in the hepatologist group had more advanced liver disease at baseline,

higher proportions of CTP class C and HCC stage C. However, improvements in QoL remained significant after adjustment for baseline QoL, suggesting that the observed benefits were robust even in the context of greater disease severity.

Third, the participating centers were more likely to be interested in palliative care integration into their practice, which may limit generalizability. However, all sites agreed to be randomized before launch of trial and followed the assigned care model, ie, those randomized to hepatologist group underwent primary palliative care training; while those randomized to palliative care group, optimized their workflows for palliative care professionals to deliver intervention.

Conclusion

This study addresses a critical gap in care of people living with ESLD, a population that has historically received palliative care only during terminal hospitalizations. Prior evidence was largely limited to retrospective observational studies highlighting underutilization of palliative care, with few interventional trials,⁵⁷ and no randomized trials comparing palliative care delivered by trained hepatologists with palliative care specialists. Given the well-documented decline in QoL as liver disease progresses,⁴²⁻⁴⁵ the magnitude and consistency of improvement in QoL observed at 3 months underscore the clinical value of earlier palliative care integration for selected patients. However, differential recruitment highlights implementation challenges. Palliative care uniquely addresses unmet needs in ESLD, including symptom burden, depression, and distress.⁵⁸ Future efforts must prioritize the systematic integration of palliative care into routine hepatology practice while offering a choice of evidence-based models tailored to local clinical and resource contexts.

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